



Analysing cross-border logistics operations for performance improvement: development and validation of a reference model

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Abstract

The critical role played by cross-border logistics (CBL) systems in sustaining fast, efficient and responsive global supply chains is widely recognised. However, scholarly research focusing on CBL systems is limited, which means that theoretical frameworks and methodological approaches available for analysing and improving CBL operations are underdeveloped. As such, coordination among diverse stakeholders, meeting regulatory requirements, adapting to a rapidly evolving technological environment, and limited capacity for investing in high-value assets, all remain ongoing challenges hindering the smooth and swift flow of cargo across borders. With a view to addressing these challenges and research gaps, this paper develops a reference model for analysing and improving CBL-specific processes. Seven underlying process dimensions have been identified following a rigorous methodological approach for the modelling, analysis and improvement of CBL operations. The efficacy of the reference model is demonstrated using two purposive case applications. The reference model also helps identify potential technological interventions to improve operations and benchmark CBL systems, operations and processes.

Keywords Process analysis and improvement · Operations improvement · Cross-border logistics · Maritime logistics · Port operations

1 Introduction

Cross-border logistics (CBL) involves the movement of cargo across national and state borders while complying with the regulations of associated territories and protocols of international trade. Significant improvements in cross-border logistics (CBL) performance have been witnessed since the introduction of standardized containers, which can

be transported on ocean vessels, trucks and trains, and transferred at intermediate facilities using mechanized equipment (Levinson 2016). Containerization has also improved the integrity and security of the cargo enclosed, as they are shielded from the impact of potential risks associated with handling by numerous intermediaries. Moreover, the ocean shipping segment has benefited from the scale economies achieved through ever-increasing vessel capacity (Notteboom et al. 2022). However, some scholars (e.g. Cheon et al. 2017) have highlighted the significant idle times involved in containers waiting for consolidation and collection at ports. As such, while shipping lines may achieve cost efficiencies, the system as a whole may still not be performing optimally from the perspectives of all stakeholders involved.

Governments are obliged to protect the society, environment and economy from illegal substances, invasive species and plant pathogens, as well as counterfeit and substandard goods, crossing the border. In certain jurisdictions, governments also generate substantial revenue in the form of taxes collected at the border (Ferrantino 2012). Operationalising these objectives creates friction across the different parts of the CBL system, leading to a slow and uneven flow

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of cargo. The literature highlights that smooth CBL flows lead to enhanced international trade, higher foreign direct investment, and economic growth (Mann 2012; Nguyen et al. 2016). Removing the hurdles at the border has also been pursued by international bodies such as the World Trade Organization (WTO) and World Customs Organization (WCO) through initiatives such as the Trade Facilitation Agreement (TFA) (WCO 2016; WTO 2014). Their proposals, such as implementing National Single Windows (NSW), are however multi-year, multi-stakeholder initiatives that not only require technical support and guidance but also substantial financial investments, which may not be readily available in all countries. Often, such high-level solutions are promoted with limited guidance on how to implement them. As such, overall, achieving an appropriate balance between the enforcement of regulatory measures and the smooth flow of goods through the border is an ongoing challenge that requires detailed investigation into the associated operations.

Extant research on improving CBL operations has predominantly focused on finding solutions to specific problems faced within the individual segments rather than the entire CBL system. For example, at the core of CBL systems, ports have been the primary focus of extant research (Garrido Salsas et al. 2022; Ha et al. 2019; Min 2022). Some well-defined problems, mostly within the port, such as berth allocation, quay crane allocation and yard truck routing, are well-suited to numerical optimisations following the positivist tradition of research (Mar-Ortiz et al. 2020; Roy and de Koster 2018; Steenken et al. 2004). However, when the scope of the investigation is broadened to include diverse stakeholders, such as customs, quarantine and hinterland transport, as well as their interactions, the ambiguity of associated problems increases significantly, thus demanding alternative approaches. In order to address these problems, the research reported in this paper develops a reference model capable of optimising CBL operations as applicable to containerised maritime cargo, to deliver system-wide performance improvements. The proposed model also seeks to identify the most impactful investments, both in terms of infrastructure and technology.

To this end, the paper aims to answer two key research questions: (i) what are the best approaches for diagnosing underlying problems in CBL; and (ii) how to improve the performance of a given CBL system. The following section of the paper examines the extant literature on CBL operations, analysis and improvement. The methodology applied in the study is presented in the next section, followed by the details of the reference model developed in the study. The application of the reference model to analyse and identify improvement opportunities in two contrasting case studies is demonstrated next. The discussion that follows reflects on

the contribution of the reference model and its limitations in light of comparable practices. While all CBL systems exhibit the characteristics described herein, the scope of the study is confined to maritime CBL systems for handling containerized cargo.

2 Literature Review

This section presents a synthesis of the literature review undertaken to establish the state-of-the-start in CBL systems and process analysis and improvement. The section also establishes the research gaps the study seeks to address.

2.1 Challenges in analysing and improving CBL operations

Based on the review of extant studies on containerized CBL operations, Fransoo and Lee (2013) have highlighted the concentration of research on container terminals from an operations research perspective, on the one hand, and aggregate trends studied from an economic perspective, on the other hand. Their study further reveals the problems faced by shippers and consignees due to unreliable and long lead times, which are in turn attributed to the focus of both shipping lines and container terminals on capacity utilization of capital-intensive assets. Such behaviour leads to buffering of containers at ports well in advance of vessel departure leading to long storage periods. Such excessive lead times and failure to provide compensation for delays incurred by the carriers have led consignees to build excessive safety stocks which in turn reduce their incentive to clear cargo immediately upon arrival further increasing the storage time. The study highlights the need for a holistic view of the CBL system that accounts for the perspectives of all stakeholders, which goes beyond the performance metrics of a single stakeholder.

Congestion within and in the vicinity of ports, caused by trucks is a well-recognized problem that adds to the transit times of cargo and escalates costs (Van Der Horst and De Langen 2008). Truck Appointment Systems (TAS) along with technologies such as smart gate systems have been implemented as solutions in some CBL systems (Caballini et al. 2020). However, they have the potential to add further burdens to users such as additional information systems, bookings that must be made well in advance, usage charges, and penalty fees for missed appointments (ACCC 2021). Therefore, such systems not only need to be well integrated into existing information systems but also regulated against cost escalations. This requires not only a holistic understanding of the CBL system for envisioning the solutions but also a detailed view of implications for end users.

Dry ports, connected to seaports by railways and situated in hinterlands have emerged as workarounds to minimise emissions and congestion within ports (Baccelli and Morino 2020). Ports also use these multi-modal hubs as means of penetrating the hinterland, especially when there is competition from multiple ports (Notteboom and Rodrigue 2005). The establishment of dry ports can be expensive considering the investment in handling equipment and railway infrastructure. Operational costs such as double handling of cargo, the need for onsite regulatory clearance, and the lack of sufficient and consistent volumes of cargo means they often need to be supported by subsidies (Baccelli and Morino 2020). Evaluation of such expensive solutions prior to implementation needs to be based on systematic investigations that use empirical data. However, such an approach considering all aspects of CBL is currently not available.

Meeting the diverse expectations of a multitude of stakeholders and the lack of effective coordination among them can be a substantial problem in CBL. Grainger and Morini (2019) have conducted a comparative investigation of interactions between businesses and regulatory agencies in CBL in Brazil and the UK to evaluate the impact on trade facilitation. The study postulates that the performance of frequent transactional interactions can be enhanced by improving the longer-term partnerships among the organizations. In the case of customs and consignee interactions, this means evaluating the regulatory compliance of consignees more broadly by validating their internal processes rather than examining individual transactions. Scholars also discuss the need for ongoing engagement of stakeholders to improve policies enhancing the environment for interactions. In addition to highlighting the location-specific differences in regulations and performance in CBL between Brazil and the UK, this study also demonstrates how poorly implemented interfaces cause friction and inflate transaction costs. However, the effect of poor interfaces in CBL extends well beyond those between consignees and regulatory agencies as discussed by the scholars. As such, interactions among regulators and logistical service providers and logistical service providers and consignees/shippers need to be analysed and improvements must be discussed, as well, for system-wide improvements.

Keceli et al. (2011) have studied the port operations in Turkey and have noted the weak control, duplication of effort and other communication problems resulting from paper-based processes and standalone information systems. Having conducted a SWOT analysis of the prevailing situation they have proposed the implementation of a Port Community System (PCS) to alleviate the problems. However, they note the implementation difficulties encountered elsewhere due to the large investments required and the challenges associated with coordination among stakeholders.

Similar problems in implementing inter-organisational information systems have been recorded in other studies, as well (Baaen et al. 2009; Tsen 2011). As such, they have proposed a phased implementation starting from resident stakeholders of the port and expanding to include others. However, their recommendations are not based on an analysis of existing operations. Adoption of inter-organisational information systems would be served well through an in-depth investigation of existing processes and their weaknesses that would aid in systems that not only preserve the prevalent best practices and knowledge while overcoming the limitations.

More recently, the application of Block chain technology has been explored to alleviate authentication and document handling problems in CBL (Wang et al. 2021). Applications of the Internet of Things (IoT) combined with machine learning and artificial intelligence to improve operations and application of digital twins to container terminals are other areas that have seen interest (Filom et al. 2022; Hofmann and Branding 2019). Despite these efforts, the level of automation and adoption of technology in CBL has lagged behind other industries (Brunila et al. 2021; ITF 2021). With the expanding basket of technologies, it has also become increasingly difficult to select the appropriate solutions to match with available funds while attaining system-wide improvements. This highlights the need for a systematic approach to identify areas that need technological interventions that could yield the highest impact, given the limited investment funds available.

Esser et al. (2020) have conducted a case study of the Port of Antwerp based on literature and interviews to understand the changing job profiles in the port sector. The study points to the decline of jobs that comprise routine tasks and jobs of an administrative nature along with the emergence of jobs assisted by emerging technologies and jobs where a multi-skilled workforce is required. As such, holistic investigations into CBL systems being socio-technical systems need to be investigated with sufficient attention paid to the human impact on these systems.

This sub-section highlighted that while there is a need for approaches that are capable of examining CBL operations holistically covering the entire cargo journey, inclusive of logistical and regulatory perspectives, there is a dearth of such approaches. Furthermore, while macro-level analysis such as capacity optimisations and IoT applications have been discussed and broad solutions such as dry ports and inter-organisational systems have been proposed, there is a need for these two perspectives to be merged at the meso-level to derive system-wide improvements.

2.2 Challenges in operations improvement

Currently, there is no dominant theoretical lens through which CBL operations can be holistically analysed and improved for the purpose of performance enhancement. In such situations, one dominant theory may need to be complemented with other relevant theoretical perspectives, considering the phenomena concerned (Halldorsson et al. 2007; Mentzer et al. 2004). CBL operations comprise a multitude of functional, organisational and contractual interfaces across which different types of cargo flow, are subject to varied protocols. Operations improvement approaches such as Lean, Six Sigma and total quality management (TQM) have all been developed based on the process view of operations. As such, the process perspective can potentially serve as a useful unifying basis for informing the analysis and improvement. However, the process view alone may not serve as a comprehensive basis for diagnosing issues and evaluating improvement opportunities. On the one hand, focusing on the process level at the expense of system-level understanding may result in issues arising out of misalignment (or conflict) of organizational goals (Ackoff 1994). On the other hand, a lack of attention to dynamic interactions at the activity level may lead to the oversight of underlying factors contributing to process issues, including human errors and procedural lapses (Samson and Bevington 2012). Therefore, there is an opportunity for complementing the process view with other theoretical perspectives such as the macro-view of stakeholders and the micro-view of role-activity interactions to compensate for the limitations of individual views. While maintaining the primary focus on the process level, zooming out can help visualize how the processes are related to the larger system (or subsystem). Similarly zooming into the processes can expose the connectivity between activities and roles with respect to decision-making. Together they allow the recognition of the influence of the contextual factors at the system level and pinpoint issues and solutions at the activity level.

As a core element of operations management frameworks such as Total Quality Management (TQM), Lean, Six Sigma, and Just-in-time, process analysis and improvement has seen a fair amount of literature over the years (Hammer and Champy 1993; Womack et al. 2007). Within the broader context of business process management, process analysis and improvement are often dealt with as an iterative or cyclic undertaking, beginning with process discovery where the “as-is” process is captured at appropriate granularity from multiple perspectives (Weske 2012). As alternative modelling notations have different capabilities with respect to capturing processes, the choice of an appropriate modelling notation is critical for process discovery and analysis. The captured process maps are then analysed to identify

various issues such as bottlenecks, redundancies, variances and inflexibilities. To help identify these issues, there is a range of tools available in the form of cause-and-effect diagrams, histograms, Pareto diagrams, scatter plots, value stream maps and process charts. However, the way these tools are used in relation to the process maps developed in the previous step is not well-articulated in the majority of extant literature. The next step, of process improvement, however, presents an even greater challenge as there is no systematic methodology or widely accepted guidelines to be used (Vanwersch et al. 2015; Vergidis et al. 2006; Zellner 2011) for addressing process issues in a methodical and holistic manner. While there are some studies that propose heuristics such as, ‘move tasks to more appropriate places’ and ‘consider if tasks can be executed in parallel’ (Mansar and Reijers 2007), their generic and subjective nature provide limited guidance in terms of achieving optimality.

Value stream mapping (VSM), seen as a vehicle to deploy Lean (Rother and Shook 2003), has been applied predominantly in analysing and improving processes in the manufacturing domain (Brunt 2000). Analogous to other improvement frameworks, modelling the “as-is” process is the starting point for a VSM exercise. While mapping the classical “as-is” process in VSM notation works well for singular flows of material, in instances of multiple flows approaching a network, the maps can become hard to comprehend and navigate, prompting alternatives such as Value Network Maps to be used (Irani and Zhou 2011). Therefore, VSM is identified as more suited for high-volume, low-mix applications (Seth et al. 2017). The subjectivity associated with classifying activities as ‘value-added’ (VA) and ‘non-value-added’ (NVA) for the purpose of removing NVA activities, has also led to the introduction of a third category of ‘business-value-added (BVA)’ activities, limiting clarity for practitioners. Within typical applications of VSM in manufacturing, transportation and storage are classified as NVA, which appears contradictory in CBL where they are important value adders. Models proposed in VSM, which capture the material flow do not sufficiently represent the nuanced flow of information and finances which can be critical in analysing CBL operations. Furthermore, centrally coordinated planning with attention to takt time, as proposed by VSM, while feasible in manufacturing, has less value in CBL operations, where decentralization is the norm.

There have been some efforts to build more fundamental theoretical perspectives to be used in the process analysis and improvement domain in particular and operations management in general. For example, the theory of constraints recommends the identification of bottlenecks in processes and resolving them by subordinating other activities to the bottleneck activity (Goldratt 1988). The approach has its merits when applied to contexts where resources are

Overall, while many approaches have been introduced to improve processes in the broader context of operations management, each with its own merits, a dominant approach aligned with the characteristics of CBL systems is yet to emerge. There is also a dearth of theoretical frameworks in the field of operations analysis and improvement. Since there is a need to improve CBL operations on a periodic basis, in the face of ongoing investments and technological advances, this research gap warrants further investigation. The study reported in this paper aimed to develop a novel reference model considering the underlying dimensions of CBL processes to identify process issues and evaluate improvement opportunities.

To address the research gaps, the study proposes to develop an empirically derived reference model to analyse and improve CBL operations based on the process perspective, while accounting for the systems perspective and preserving the activity level details. As such, the overall methodological framework depicted in Fig. 1 was adopted to inform the empirical investigation and to derive the reference model.

The paucity of holistic studies in CBL unveiled through the review of extant literature meant that documented, detailed and complete process maps were also not readily available. Therefore, capturing the operations of the selected CBL systems in process maps emerged as a primary requirement for the study. Furthermore, selecting a suitable modelling formalism to comprehensively capture CBL operations was deemed critical, as the information captured within process maps would inform subsequent analysis. Extant literature on process modelling (Weske 2019) advocates the capture of functional, organizational, informational and control (also referred to as behavioural) dimensions in a process map. The functional dimension of a process represents the (value-adding) activities of an operation. While the control dimension represents decision-making and workflow, the informational dimension represents the information used for decision-making, as well as guiding functional activities. The organizational dimension captures the entities (e.g. organizations or actors) executing the activities of the process.

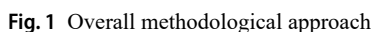
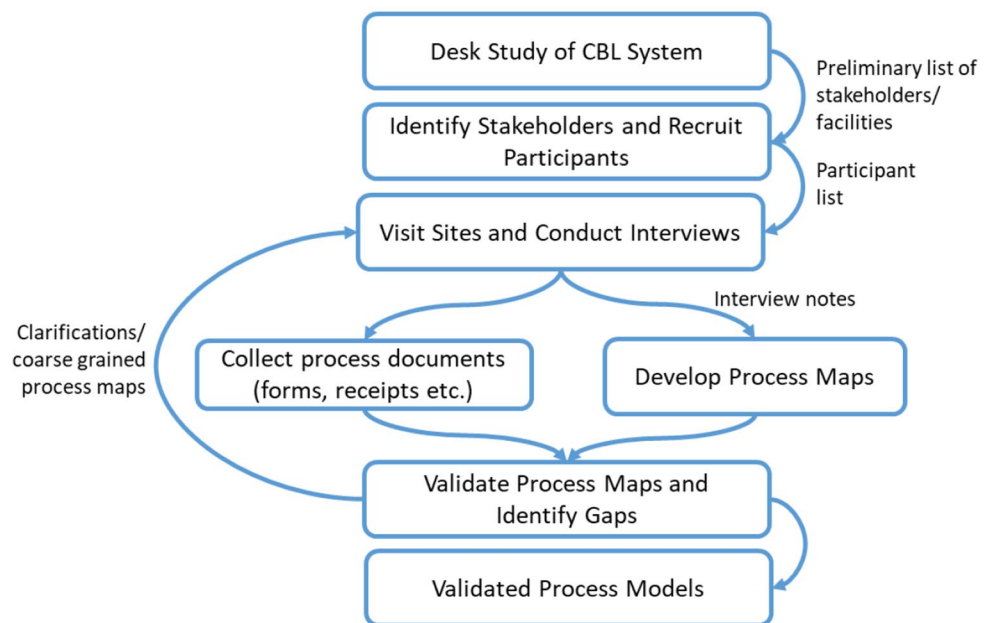


Fig. 2 Case study approach

While the above four dimensions are adequate for capturing most operations, specific domains require further perspectives for a comprehensive representation. As it is critical that movements and attributes of cargo and transporters are captured by the modelling formalism, spatial dimension emerged as essential for the comprehensive representation of a CBL system. The temporal dimension, which not only represents the queuing, waiting and execution times of the activities but also forms the basis of performance metrics such as asset utilisation was also deemed essential for analysis. The interactions among the multitude of organizational entities, a fundamental aspect of multi-stakeholder settings, which cause multiple process issues need to be captured to identify and analyse potential operational issues caused at these interfaces. Therefore, in addition to the four basic dimensions, spatial, temporal and interactional dimensions were determined to be essential to capture the CBL processes holistically for analysis and improvement. An evaluation of widely used process mapping formalisms revealed that Role Activity Diagrams (RAD) are capable of capturing five of the requisite dimensions (the highest number of any individual formalism available) natively while the complementary dimensions (spatial and temporal) could be integrated as addendums without sacrificing the ease of comprehension (Bandaranayake, Kiridena, & Kulatunga, 2022b). Thus, RAD with extensions for spatial and temporal dimensions was used for capturing the processes.

Often, case studies are considered to be most appropriate for inductive, theory-building research (Gammelgaard 2017; Kiridena and Fitzgerald 2006) and are capable of providing rich insights into the influence of contextual factors (Yin 2014). There are significant numbers of research in the

CBL domain, especially those focused on ports, have followed a case-based approach, presumably due to the diversity of CBL operations in different geographical locations. Thus, the study adopted a case-based approach for empirical data collection, given that the objective was to develop a novel reference model. The two selected case studies revealed the unique and context-dependent aspects of each CBL system while laying the foundation for the development of a context-independent reference model.

CBL operations through the Port of Colombo in Sri Lanka were the subject of the first case study. The Port of Colombo is a major hub port in the midst of east-west shipping lanes, facilitating the transshipment of containers, and ranks among the top 30 global ports in terms of the number of containers handled (ADB 2020). The second case study location was Australia, where processes through several container ports, primarily dedicated to the import and export of cargo, were studied. The differences in geographies, modes of operation (hub vs. gateway), and the degree of digitalization/automation of the two contrasting case studies provided a rich set of data for further analysis of CBL processes.

The approach followed in conducting the case studies is depicted in Fig. 2. In the first phase of process discovery, documented literature was examined, the key stakeholders were identified, and preliminary process maps were created. However, the development and verification of full-blown process maps heavily relied on interviews conducted with a range of stakeholders iteratively. The coarse-grained process maps developed through literature were used to guide the interviews to gather further details of processes. The information from interviews, process documents and insights from observations were used to develop granular

Table 1 Details of interviews conducted

Participant Organisation	Sri Lanka		Australia	
	Number of Interviews	Duration/ (hours)	Number of Interviews	Duration/ (hours)
Customs brokers & freight forwarders	32	44	5	5
Cargo processing centres	3	4	2	3
Government agencies	20	26	-	-
Port authority	12	18	2	2
Shippers and consignees	3	4	1	1
Shipping lines	4	6	1	2
Terminal operators	2	3	4	8
Trade bodies	3	3	-	-
Total	79	107	16	21

Table 2 Process maps developed in the study

	Sri Lanka	Australia	Total
Process maps for import cargo	48	29	77
Process maps for export cargo	39	21	60
Total	87	50	137

process maps which were verified and refined during subsequent interviews. As depicted in Table 1, the total duration of interviews during the study period of two years amounted to more than 100 h. The process discovery resulted in a comprehensive set of 140 process maps, arranged in three levels of granularity depicting not only high-level operations but also granular details of processes for each location. The process maps for import cargo encompass details from the point of reporting the cargo manifest by the shipping line to the customs authority until the cargo is unloaded, cleared and delivered to the consignee. The process maps for export cargo cover the details from the point of making a booking with the shipping line until, export clearances are obtained, cargo is loaded onto the vessel and the vessel departs the port. A summary of the process maps developed are shown in Table 2.

The validity of data gathered through interviews is contingent upon the participants' perception, their ability to describe the operations and the interviewer's interpretation of the descriptions. These threats to validity were mitigated by engaging multiple participants from the same stakeholder group and validating the documented process maps during the next iteration of the interview process. The developed process maps were also validated through several workshops and, to a limited extent, via process mining (van der Aalst 2016) by analysing information system logs obtained from the customs and port information systems. Apart from the interviews, during the latter stages, clarifications regarding some of the processes were obtained by email, as well.

The aim of the next stage was to develop an analysis approach that extended naturally from the process maps (in-vivo) rather than having to employ a disparate set of tools. The seven dimensions adopted as the foundation for the modelling formalism were deemed appropriate for guiding

the analysis and improvement, as well. Activities (A), technologies (T) and properties (P), of each process dimension, discovered through a detailed examination of process maps, were used to further characterise each dimension. Activities represent the value-adding steps in each dimension that require effort. Technologies used in executing those activities identify the maturity of the dimension in terms of improvements deployed. Properties capture the attributes that further describe the dimensions that are not captured by activities and technologies.

The reference model was derived to represent the flow of information, one of the process dimensions, across the other process dimensions by examining the seven process dimensions, the attributes and the relationships among the dimensions. As such, the model is able to identify and explain the underlying performance drivers of the processes. The process maps were partitioned for validation, analogous to the approach used in machine learning, where similar processes with contrasting performance across the two cases were identified. Processes with lower performance were traced through the reference model to identify the weaknesses and improvement opportunities. Thereafter the same approach was followed with higher-performing processes to verify those process issues have indeed been resolved. This provided the validation that the reference model is capable of highlighting process issues and identifying improvement opportunities.

With the advent of novel technologies such as the Internet of Things (IoT) and cloud computing, a new suite of technologies has emerged, commonly referred to as Industry 4.0, that has promised to improve process performance further. However, incorporating such technologies into operations in a manner that improves performance requires a deeper appreciation of the characteristics of the technologies. As such, an additional literature review was undertaken (step 6 in Fig. 1) to establish the state-of-the-art in those technologies as applicable to CBL, with a view to assessing their suitability during the improvement stage (Bandaranayake, Kiridena, & Kulatunga, 2022a). Those identified technologies were then mapped onto the technology attribute of each

process dimension of the reference model so they could be used in the improvement phase.

This section describes the overall methodological approach for the study starting from the derivation of the process mapping formalism to the conceptualisation of the reference model and its validation. The next section elaborates on the proposed reference model and its constituent elements further.

4 Reference model for CBL processes

Figure 3 represents the reference model developed in the study where a given CBL operation is viewed as a system. The connectivity of the (sub) system to the external environment is represented by an arrow through the organizational boundary extending from the interactional dimension. The four dimensions within the shaded area comprise activities, that require effort, skills, knowledge and capacity of the organizational entities/actors and consume time. Those entities and actors are captured within the organizational dimension while the temporal dimension captures various times associated with the processes involved. These two dimensions are represented outside the shaded area to depict their applicability/relationship to all other dimensions. The information dimension represents feed-forward and feedback loops connecting all dimensions. The numbers 1 to 7 denote the proposed sequence of analysis when using the reference model.

Each dimension of the reference model is depicted graphically (Figs. 4, 5, 6, 7 and 8) and briefly discussed in

the remainder of this section. In addition to the attributes of each dimension, inputs and outputs depict their connectivity to other dimensions.

4.1 Control dimension

As shown in Fig. 4, the control dimension represents the decision points in CBL systems that result in alternate courses of action, provision of execution parameters (position/direction etc.), acquisition of resources and triggering of activities. The activities in the functional dimension prepare and present a ‘decision frame’ to the control dimension to perform control actions, which includes objectives, decision variables, decision criteria and constraints applicable to each decision.

4.2 Functional dimension

Activities in the functional dimension are sub-divided into spatial and informational groups as elaborated in Fig. 5. Physical activities impacting the spatial dimension enable the movement of cargo, adding place utility to cargo. The informational activities process information acquired from spatial and interactional dimensions into a decision frame, as an input to the control dimension. The decisions from the control dimension along with the procedures from the informational dimension aid in the execution of activities in the functional dimension.

Fig. 3 Reference model for CBL processes

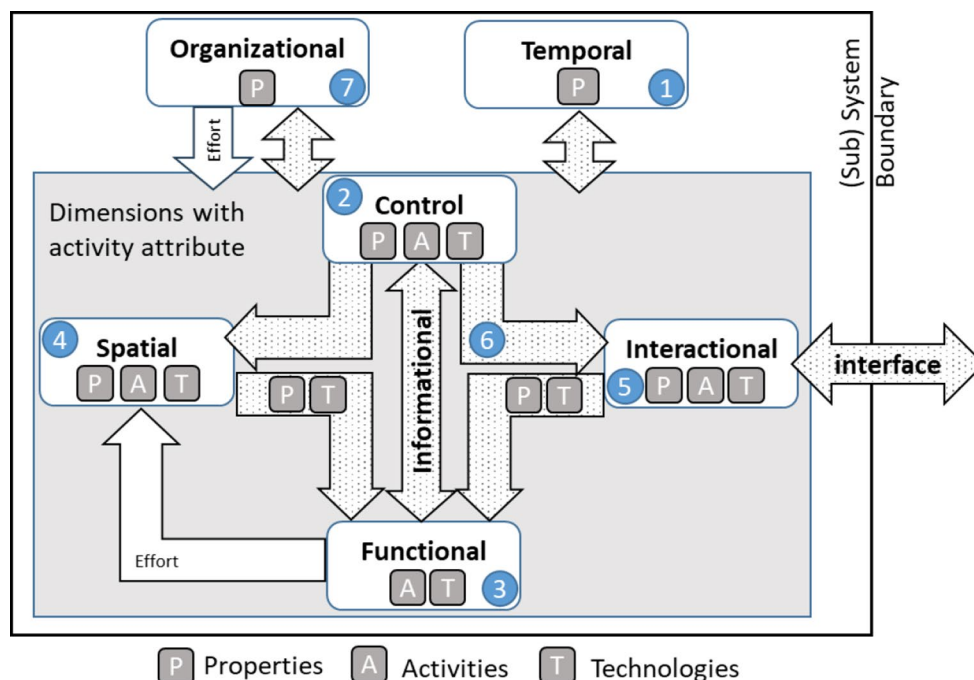
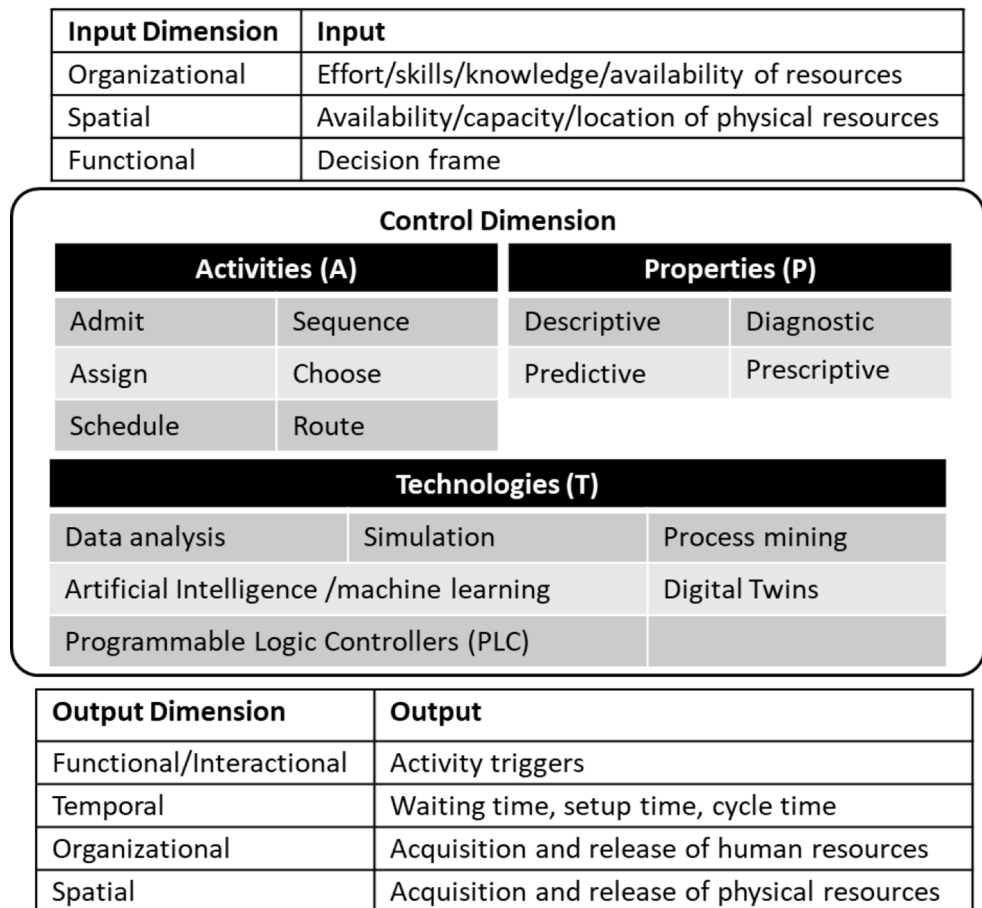


Fig. 4 Control dimension

4.3 Interactional dimension

The interactional dimension shown in Fig. 6 represents the communication interface of the (sub) system under analysis with the outside world. As information flows outside of the internal controls of an organizational entity, confidentiality and integrity of information become significant considerations. As such, the activities in the dimension represent those associated with sending and receiving information with appropriate security protocols in place. Associated technologies include those that support active and passive information sharing. Compared to the spatial dimension from where information is often acquired in real-time, information requirements from external stakeholders could be obtained in advance and buffered to avoid delays in execution.

4.4 Spatial dimension

Capturing of dynamic and static properties of physical objects represented by the spatial dimension is elaborated in Fig. 7. Dynamic properties that change in the short term due to physical activities can be either captured manually and/or

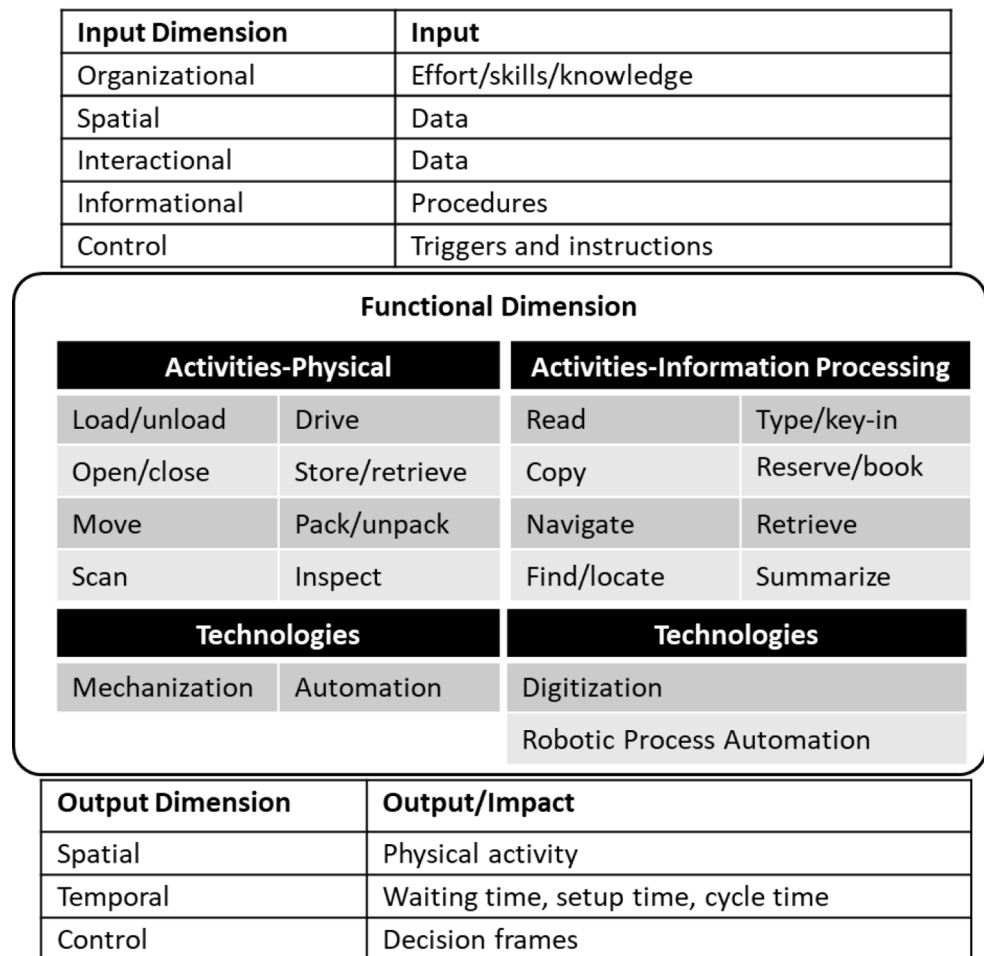
using technologies ranging from handheld devices to autonomous sensors with IOT functionality. In contrast, the static properties of the assets, once acquired can be stored until those assets are replaced or modified. In addition to their use in decision-making and execution, information captured from the spatial dimension lays the foundation for tracking and tracing, a key requirement in logistics.

4.5 Informational, organizational and temporal dimensions

Non-activity type dimensions of the reference model are depicted in Fig. 8. The informational dimension acts as a medium that connects all the dimensions and also represents the accumulated knowledge within the organization in terms of procedures, manuals and other structured information. Applicable technologies are categorized into those related to the storage and communication of information.

The temporal dimension captures parameters that track the temporal performance of the sub-system or the processes under review. In addition to the primary temporal measures, derived measures such as utilization are also captured within this dimension. The organizational dimension captures the

Fig. 5 Functional dimension



properties of the human resources, which not only enable the activities but also act to improve the processes continuously. Human resources hold the tacit knowledge of the organization that underpins organizational capabilities.

5 Data Analysis and results

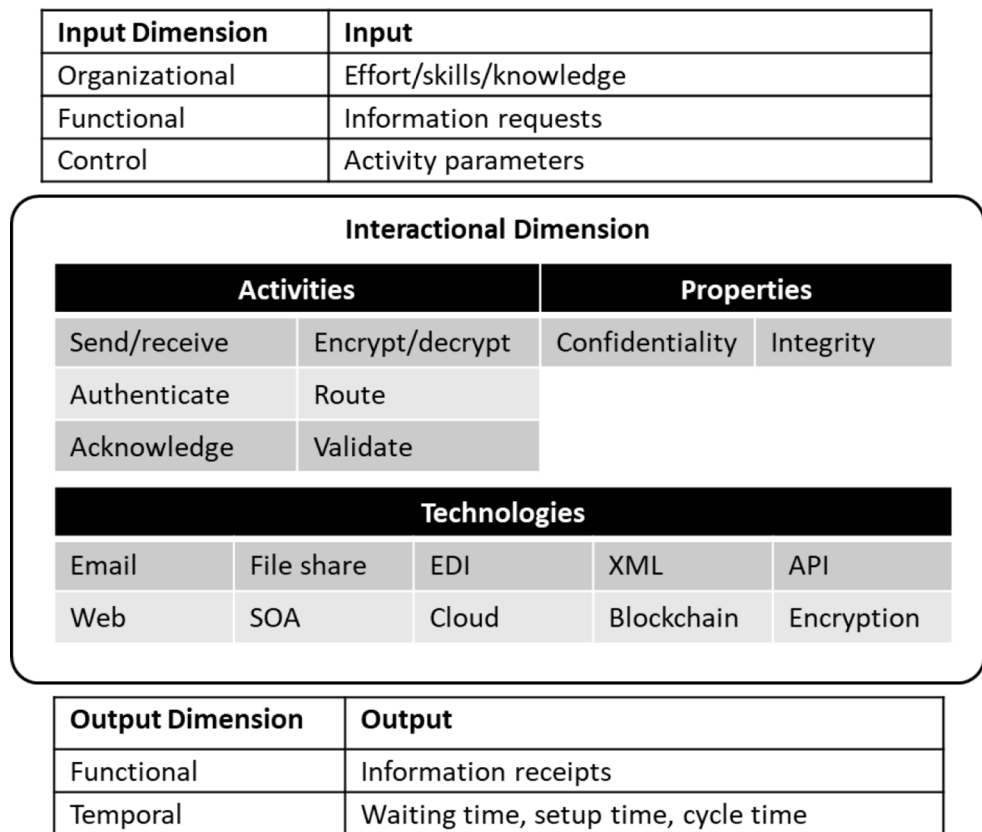
This section demonstrates the analysis and improvement of several different operations in CBL using the reference model at different levels of detail. First, the receipt of export containers at the container terminal is examined at the aggregate level, followed by a detailed analysis of the container gate-in process in both Australian and Sri Lankan CBL systems. Next, the assessment and release of cargo by customs is examined at the aggregate level followed by a detailed analysis of the assessment of customs declaration. Finally, cargo preparation at the intermodal terminal is examined in detail, which is only applicable to the Australian system. These operations have been selected to demonstrate the analysis and improvement capabilities of

the reference model both at the different levels of detail in comparative and stand-alone modes.

5.1 Receipt of export containers at the container terminal

Receipt of containers from the hinterland into a container terminal can affect yard crane productivity, gate utilisation and formation of queues within and in the vicinity of the port. This operation consists of several processes; receiving information on incoming containers; admitting trucks into the yard and unloading containers into the yard. The activities and systems supporting them for each of the dimensions of the operation are shown in Table 3. The process of admitting trucks into the terminal is further elaborated in Fig. 9 highlighting the dimensions of the reference model.

The analysis is initiated in the temporal dimension where truck turnaround time, is one of the primary performance drivers. The truck turnaround time has been determined as 3 h in Sri Lanka (Rathnayake et al. 2015) and 0.5 h in Australia (BITRE 2021) which shows a sharp contrast. The control decision of admitting trucks into the yard (further

Fig. 6 Interactional dimension

elaborated in the process map in Fig. 9), compares information from several external (interactional) and internal (spatial) sources, to ensure only containers approved by shipping lines and regulators are received during the vessel opening window. The preparation of the decision frame is captured in the functional dimension where information from different sources is combined and presented in a suitable format for a prompt decision.

The truck admission decision when presented in digital format, can be automated as implemented in the Australian system improving the processing time substantially. When information is acquired in digitised format as in the Australian system and fed into the information system of the container terminal, the functional activities of preparing decision frames are minimized presenting another set of improvement opportunities. While the information from the spatial dimension (containers and trucks) must be obtained in real-time, the information from the interactive dimension can be obtained in advance and buffered to avoid delays. The third-party message brokering system implemented in the Australian system enhances activities in the interactional dimension substantially by eliminating the need for pairwise interfacing of information systems. The message broker has implemented a single interface with each stakeholder's information system and acts as a conduit carrying messages among them. This message brokering system

eases the burden on consignees to submit the same information to the information systems of multiple stakeholders as observed in Sri Lanka.

5.2 Container gate-in process

Since truck arrival at the gate typically follows a stochastic distribution, swift admission alone cannot avoid congestion during a surge in arrivals. However, the truck arrival can be evenly distributed throughout the day via the implementation of a truck appointment system (TAS). Two process maps shown in Fig. 9 correspond to the gate-in process for export containers at the Port of Colombo (Fig. 9 (a)), where the process is not digitalized and at Australian Ports (Fig. 9 (b)) where the process is digitalized to a substantial extent along with a TAS.

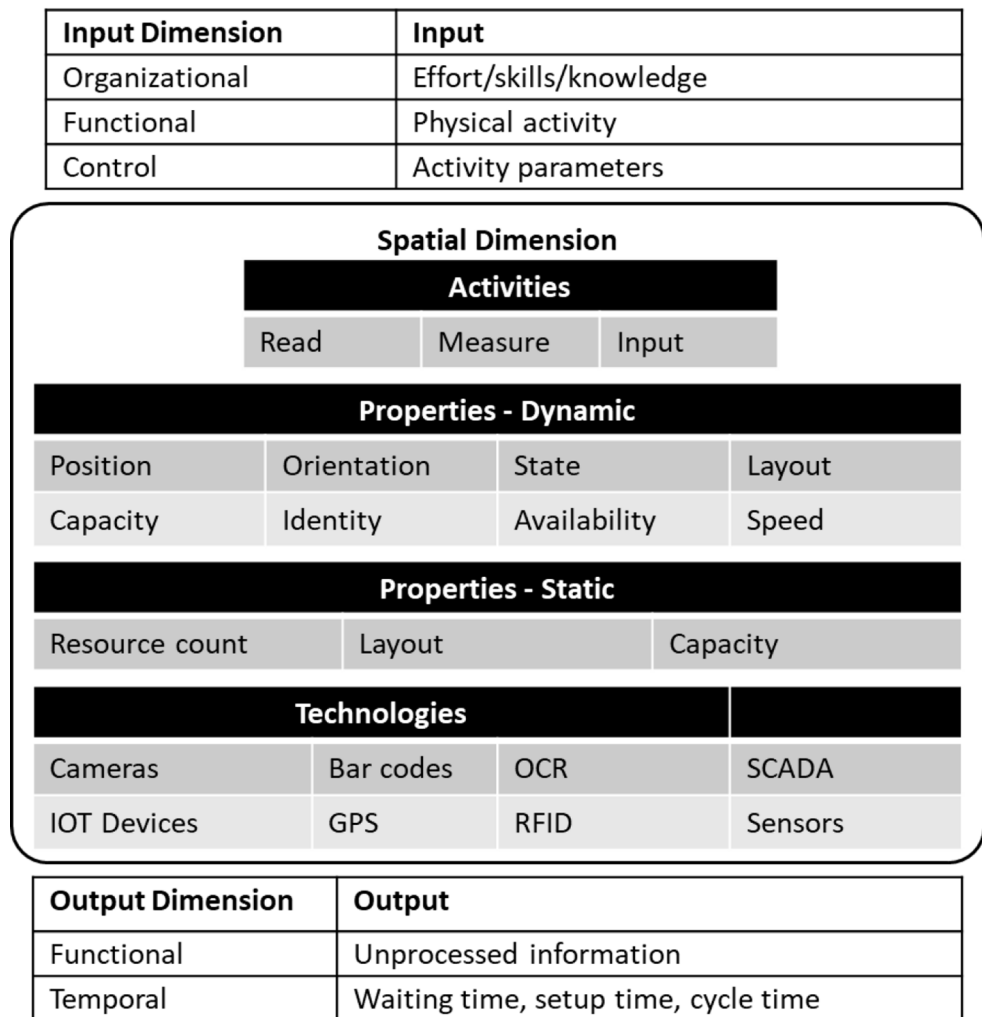
Due to the presence of an appointment system and smart gate systems at Australian ports.

$$\sum T_x(a) > \sum T_x(b)$$

Where $T_x(y)$ is the activity time for segment x shown in Fig. 9(y).

However, this gain in performance comes at an additional cost to the trucking company, which must reserve a time slot before arriving at the port gate, which would also

Fig. 7 Spatial dimension



mandate IT infrastructure and organizational training. Such an impact on the organizational dimension could pose challenges to trucking companies that may not be technically savvy. However, the cost could be offset through the reduction of waiting time at the gate which reduces costs in terms of fuel and payments to the crew. From the terminal's perspective, the additional cost of a TAS helps to balance the gate and yard crane utilization and in the Australian experience, additional revenue for the use of the reservation system as well.

$$T2(a) \gg T2(b)$$

The time-saving in segment T2 in the Australian system is achieved in the spatial dimension where the Optical Character Recognition (OCR) system acquires the truck and container numbers. The implementation of spatial information acquisition technology at the gate should be evaluated by weighing the investment against the prevailing labour costs

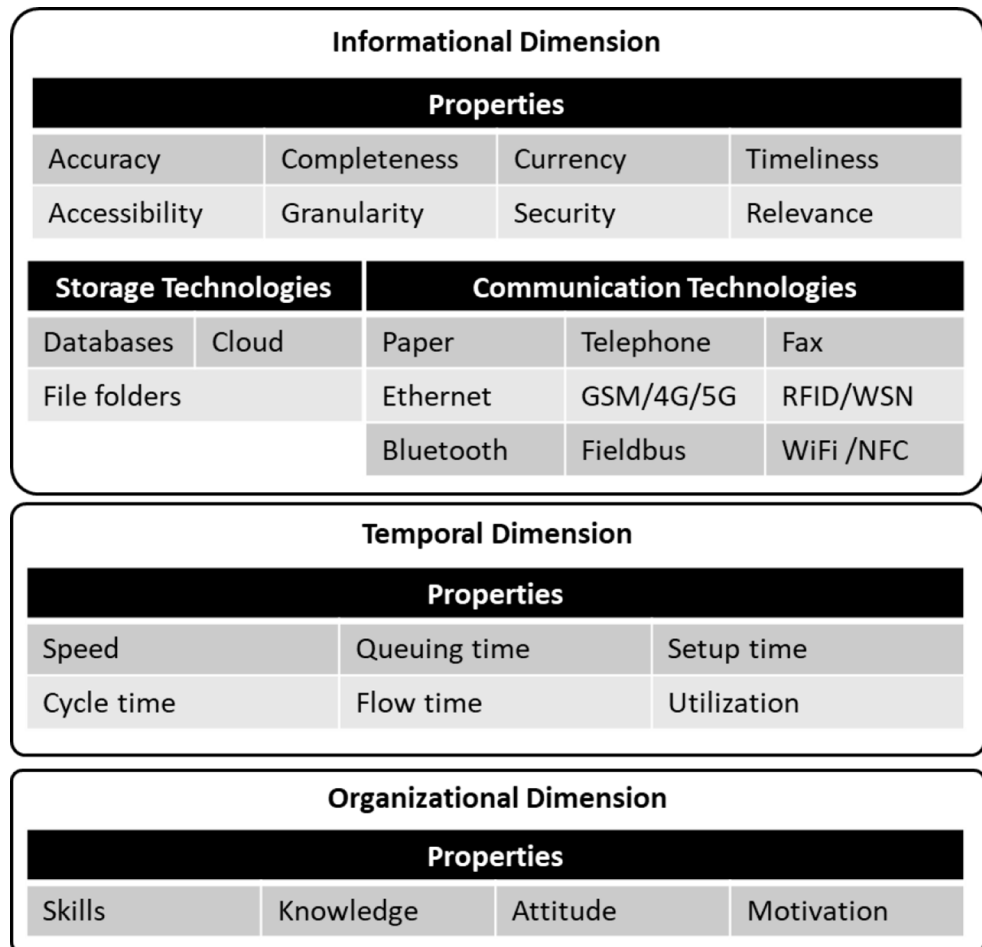
and congestion at the gate. Even in Australian CBL systems OCR equipment is only installed in selected terminals.

$$T4(a) \gg T4(b)$$

The improvement of segment T4 shows the synergy of the TAS and the digital information acquisition systems where matching activity is performed by computers (control dimension). The insight from this analysis is that once information is acquired in the desired format (digitized truck, container ID and driver ID in this instance) additional functional activities such as keying in information are also eliminated.

$$T3(a) > T3(b)$$

Since an RFID-enabled driver ID is read faster and eliminates manual comparison, time is saved in segment T3 as well. While unlikely, the step is prone to mishandling since driver ID could be carried by anyone. Ideal authentication

Fig. 8 Informational, organizational and temporal dimensions

of the driver should be based on biometrics and without the driver having to reach out of the vehicle, to further improve speed, integrity and safety.

In addition to the improvements discussed above, the acquisition of information in digitized form may help in downstream processes, as well. For example, the gate-in status can be communicated to the container yard immediately for preparatory activities.

5.3 Assessment and release of cargo by customs

Customs organizations are mandated to ensure that the cargo crossing the border is compliant with the legislation and that taxes due are collected. These objectives are accomplished through the analysis of information pertaining to cargo, received from the consignee and the carrier. However, if the customs agency is not satisfied with the validity of the information received, or if the information raises concerns, the cargo may be scanned or inspected. The activities of the operation and systems supporting the activities are listed in Table 4. The main performance metric for this operation is the release time, defined as the time taken to release cargo from the customs purview. In Australia, the release time is

2.4 h from the time of cargo arriving at the port (DHA 2016). In Sri Lanka, the latest time release study (TRS) shows the release time as 16.5 h, although it is measured from the time of filing a customs declaration (SLC 2019). Data gathered during the study shows that on average it takes 3 days to file the customs declaration from the time of cargo arrival at the port. As such the release times are significantly different for the two CBL systems.

The decision to release cargo involves validating and comparing the declaration and other supporting documents from the consignee and the cargo manifest from the carrier. Other than information obtained through the interactional dimension, the decision is also supported by the risk profile for the shipment. The risk profiles are generated from historical data for the commodity type, country of origin and consignee, with each profile assigned a level of risk based on compliance history. As such, the control decision of flagging high-risk cargo can be automated if all relevant information is digitized. However, due to difficulties in validating the authenticity of documents and whether the documents fairly represent the cargo, human judgment is often applied to decide whether to release the cargo or to inspect it. The operation relies heavily on information obtained from the

Table 3 Port sub system: acceptance of cargo from Hinterland

Dimension	Activities	Systems or People Executing/Supporting the Activities	
		Sri Lankan System	Australian System
Control	Admit authorized vehicles and approved containers into the terminal	Gatekeepers	Smart gate system supported by truck appointment system (TAS)
	Determine the stacking position for unloaded containers	Supported by terminal operating system (TOS)	
Spatial	Allocate cranes for unloading		
	Read truck, trailer and container numbers	Gatekeepers	Smart gate system (cameras and sensors)
	Read driver ID	Gatekeepers	RFID readers
	Acquire yard capacity and current space availability	Records of the previous movement	Automated handling system (cameras and sensors)
Interactional	Acquire yard crane status	Verbal communication with crane operators	Automated handling system
	Receive reservation details from shipping lines (vessel, shipper, container IDs, destination)	Batch files from the shipping line	
	Receive vessel arrival time from the shipping line		
	Receive pre-receipt advice (PRA) from consignees (container ID, destination, vessel, weight)	Web interface to TOS	Third-party message brokering system
	Receive clearance to accept/load cargo from the regulator	System interface to regulator's system	
	Provide cargo receipt window to consignees	Website of terminal	
Functional	Provide cargo receipt notice to the regulator	System interface to regulator's system	
	Frame information for the admission decision	Gatekeepers	Smart gate system and TAS
	Open gate for trucks	Gatekeepers	Smart gate system
	Frame information for unloading location decision	Supported by TOS	
	Unload containers from trucks onto the yard	Crane operators	Automated handling system

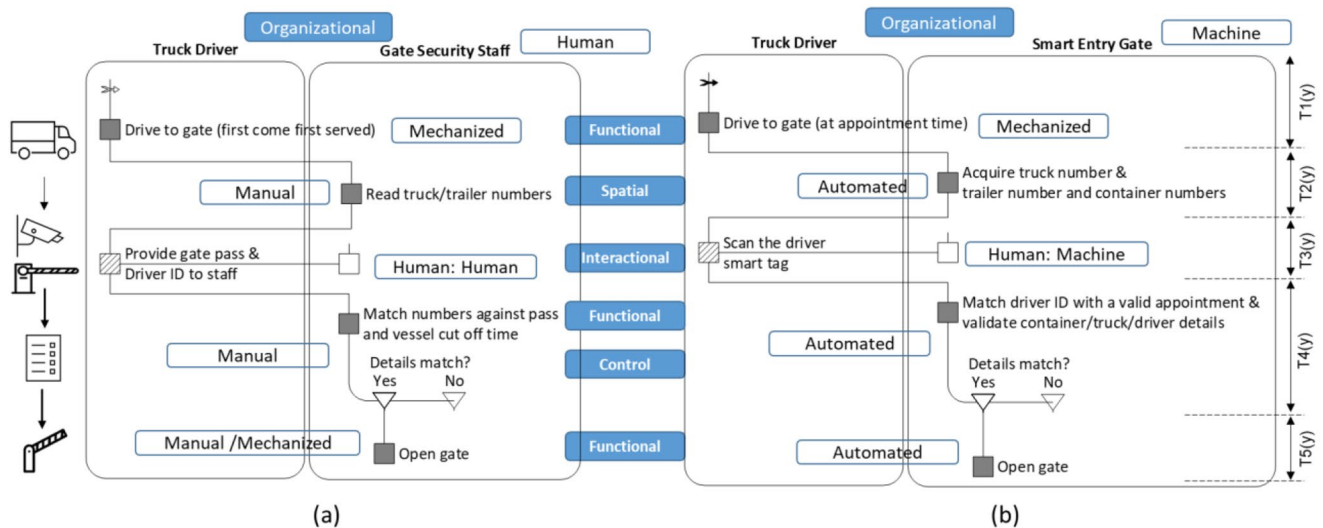
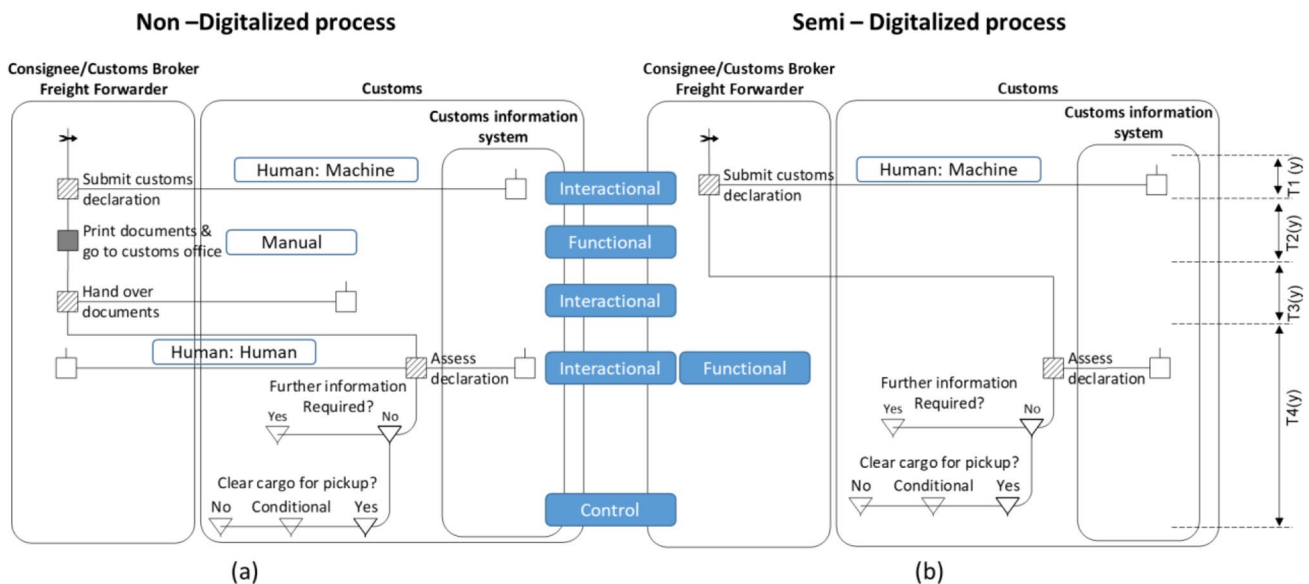
**Fig. 9** Analysis of Gate-in Process for Export Containers

Table 4 Assessment of cargo declarations

Dimension	Activities	Systems or People Executing/Supporting the Activities		
		Sri Lankan System	Australian System	
Control	Release cargo from the customs purview	Customs officers supported by customs information system in the presence of customs house agents (CHA)	Customs officers supported by the customs information system	
	Flag high-risk shipments			
	Determine the validity of declared classification and value			
	Determine the authenticity and accuracy of information received (risk)			
	Determine the tax revenue due			
Spatial	Verify the receipt of taxes	Customs information system	Customs information system	
	Verify the quantity of cargo (weight, volume, unit count)	Customs officers	Customs officers	
	Verify the cargo description/classification			
	Determine the fitness for use			
Interactional	Receive cargo manifest from the carrier	XML files from shipping lines	Interfaces to client systems via application programming interfaces (API)	
	Receive the customs declaration from the consignee	Web interface to customs information system and printed copies carried by CHA		
	Receive supporting information from the consignee	Customs information system		
	Provide tax computation to the consignee	System interface to terminal operator's system		Third-party message brokering system
	Provide release status to the consignee and shipping terminal	Physical copies		
Functional	Provide declared information to other agencies	Customs information system and printed documents	Customs information system	
	Frame information for assessment	Customs information system		
	Frame information for tax computation	Cargo scanners		
	Update release status			
	Scan cargo			
	Inspect cargo			Customs officers

**Fig. 10** Assessment of customs declaration

interactional dimension. In both CBL systems information is obtained in digital format. However, in the Sri Lankan system, the customs house agent is required to go to the customs office with printed documents as well which significantly undermines the benefits of the digital system.

Obtaining information from the party that created the information would be the optimal way to verify the authenticity of the information. For example, obtaining invoices from the shipper, product specifications from the manufacturer, phytosanitary certificate from the authority in the

originating country, would be ideal. However, this would increase the number of interfaces exponentially and is impractical with the prevailing technology. The next best approach therefore could be ensuring the immutability of records from the originator. To this end, enabling technologies such as blockchains have been proposed (Ganne 2018; UNECE 2020) but mass adoption is yet to be seen.

When the information obtained through the interactional dimension is unsatisfactory, customs may resort to the acquisition of first-hand information from the spatial dimension (scanning or inspection) which adds significant functional activities such as moving cargo to a scanning or inspection facility increasing the release time. While 100% inspection is the only way to fully ensure the declared information and actual cargo are perfectly matched, it's infeasible in practice. Therefore, customs organizations resort to determining risk levels associated with declarations combined with random inspections.

5.4 Customs declaration assessment process

Figure 10 (a) depicts the process of submission of customs declaration by the consignees or their agents and the subsequent assessment by customs as practised in Sri Lanka while Fig. 10 (b) shows the process in Australia. The time from submission of the declaration to the time of release decision can be considered the performance parameters.

$$T3(a) + T4(a) \gg T4(b) + T4(b)$$

since $T3(b) = 0$ and $T4(b) = 0$

The submission of customs declaration in both instances is an online activity, where the submitter keys in the information. The activity requires substantial effort on the part of the declarant to gather information from multiple sources such as purchase orders, invoices and shipping documents. Another layer of activity is added when execution is performed by a customs broker who must obtain the information from the consignee. Electronic interfaces among information systems of stakeholders present an improvement opportunity.

The assessment of customs declaration, an interactional activity performed in the presence of the consignee or their agent in Sri Lanka, adds a substantial burden on the declarant or their agent, having to print the documents and travel to the customs office. The additional time for this is significant as well, considering the interaction can only take place during office hours. These functional activities could be substantially eliminated with the adoption of digital signatures. Calling for original documentation can be reserved for declarations deemed higher risk since risk profiling with declared digital information is possible. However,

acceptance of digital signatures may require some investments, organizational skills and attitudinal adaptations from multiple stakeholders.

5.5 Cargo preparation at intermodal terminal for delivery to port

The preparation of cargo to be delivered to the port by train is shown in Fig. 11. This process is specific to the Australian CBL system as cargo is only delivered by truck to the Port of Colombo. The cargo is accepted at an Inter-modal Terminal (IMT) and is delivered by train to the container terminal, not only reducing traffic congestion in and around the container terminal but also reducing carbon emissions, since dozens of trucks are replaced by a single train. IMT is an (additional) stakeholder, which links the shipper, container terminal and trucking company, necessitating interactive activities to ensure information flow is not disrupted.

The key activities in Fig. 11, are the loading of the train and the submission of the 'train consist', which indicates the list of containers and their positions in the train, to the container terminal. The rest of the activities are essentially error-proofing and error-correcting activities. The shipper has the responsibility to ensure that containers delivered to the IMT have been accepted by the target vessel and have been cleared by customs, which is commonly known as Pre-Receive Advising (PRA).

The control decision of selecting containers to load the train is based on the vessel departure times. This decision frame does not however consider the PRA status of each container at this stage as the system is not set up as such. Once the list of containers to be loaded on the train is decided the information is sent to the container terminal via EDI through a commercial message broker (denoted by T1 in Fig. 11). The message broker system transmits the message to the Terminal Operating System (TOS) and also validates the PRA status of each container (T2). Any errors detected remain within the message broker system, which can be checked for each container, through a human-to-machine interactive activity (T4). As it can be a costly and time-consuming operation, the planning staff at IMT only check a few randomly selected containers. If an error is detected, the IMT will attempt to remove such containers from the train if sufficient time remains for the train departure. If not unloaded, or incorrect PRA status is not detected, the container is transported to the terminal and back, which is a costly and wasteful operation. The multi-step process, generating multiple functional activities, is required as there is no exception report generation regarding the PRA status.

An activity of generating a PRA exception report, prior to loading the train, through an enhancement to the system-to-system interface could prevent the loading of the container

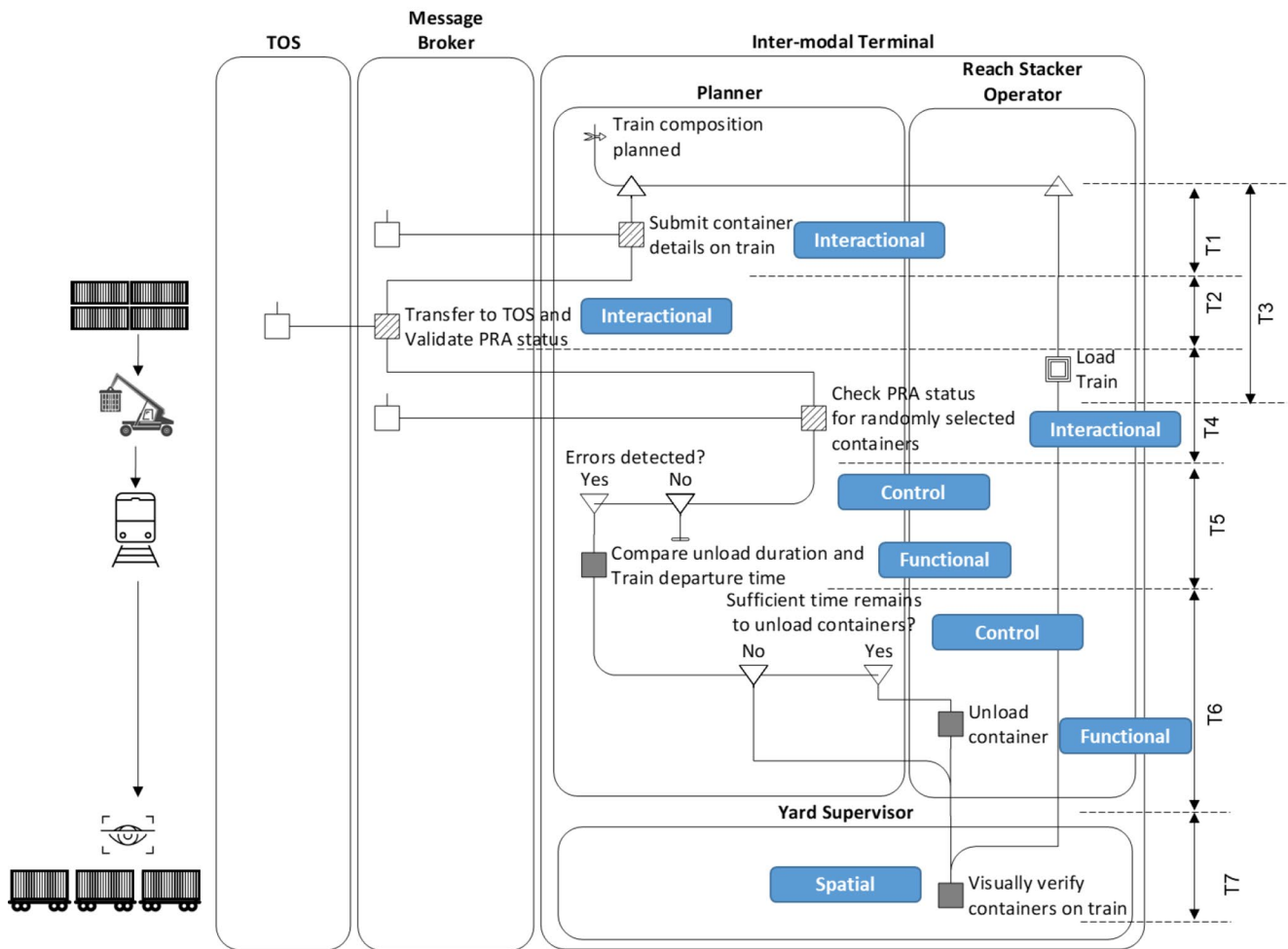


Fig. 11 Preparation of cargo for delivery at inter-modal terminal

onto the train eliminating the functional and control activities denoted by T4, T5 and T6. However, as the onus is on the shipper to verify the PRA status and the IMT can charge the shipper for any additional functional activities of handling or transportation, IMT's incentive to invest is moderated. Such improvements, therefore, require the intervention of regulatory bodies.

The last activity of the process, denoted by T6, is a spatial information acquisition activity of reading container details on the train and comparing them against the train consist. A yard supervisor moves along the length of the train reading container numbers one by one matching against the printed document. This acts as a last line of defence, to guard against any errors in loading the train. Given the length of the train (up to 900 m long), automatic data acquisition at this point would pose challenges. The train loading process (shown at the aggregate level in Fig. 11) consists of picking containers from the yard according to instructions given by the yard information system and placing them on the train. Using OCR at the point of loading through a camera mounted on

the reach stacker, which would alert any errors immediately, can be a considered feasible option, which would eliminate any subsequent verification and correction activities.

As demonstrated through the analysis, the application of the proposed reference model aids in revealing process issues and discovering improvement opportunities in CBL operations. Apart from the analysis of processes on their own merits, the potential for the reference model to benchmark processes across different systems is also demonstrated.

6 Discussion

The methodology proposed in the study required CBL systems to be represented through a set of process maps for analysis and improvement. To build such process maps, seven underlying process dimensions were identified and an RAD-based mapping formalism was developed to capture those dimensions (Bandaranayake, Kiridena, & Kulatunga, 2022b). Considering the multitude of modelling formalisms

available and the difficulty in evaluating them to identify the most suited option, the modelling formalism presented in the study is a valuable tool that can be readily applied to capture CBL processes comprehensively. The RAD-based process maps have an appropriate balance between formality and ease of comprehension, which is important in process discovery and analysis.

The comprehensive process maps developed in the study on two CBL systems are the first of their kind in terms of coverage and detail to the authors' best knowledge. Previous attempts at developing process maps were either limited to segments (Elbert et al. 2017), applicable to only a few commodities (Khumalo et al. 2023) or were only at the aggregate level (Hausman et al. 2010). Comparatively, the process maps developed in this study cover both import and export processes, apply to diverse sets of goods and are available in both aggregate and at highly granular levels. Not only do they enable analysis of the two CBL systems, but also act as a blueprint for detailed process discoveries in other CBL systems.

The reference model makes use of the information captured within the seven process dimensions, thus providing a systematic approach and a seamless path from process mapping to analysis. As such, potential improvements to the processes emerge naturally from the analysis. This is a significant improvement from the use of ad-hoc application of a multitude of tools for process analysis and improvement. The analysis takes place at the process level but also enables delving into the activity level to reveal performance drivers at the granular level. Furthermore, the analysis takes place in the context of the broader CBL system as the reference model represents CBL operations as an open system comprising inputs, outputs, feedback and feedforward loops, boundaries and interfaces. This approach leads to the identification of performance drivers and inhibitors within the system reducing opportunities for sub-optimisations within isolated segments. While some recent literature (Sarkar et al. 2022) has recognised the significance of analysing CBL operations as systems, such efforts have been hampered by a lack of detailed operational data. The detail-oriented nature of the proposed approach overcomes these limitations.

Empirical evidence gathered and analysed as part of this study suggests that more than 60% of the CBL processes comprise processing information rather than moving cargo, which is the primary objective. However, the reference model does not classify activities as 'non-value-add' if they do not add immediate value as proposed by approaches such as VSM (Rother and Shook 2003). On the contrary, leveraging the use of information in organizations such as customs could reduce a multitude of additional functional activities such as transportation, de-stuffing, inspection and re-stuffing containers. As such, the reference model offers a

novel perspective on the value of activities in the context of the system that informs process redesign.

The temporal dimension, used as the performance measure in the reference model, has been used as a performance measure in CBL elsewhere, as well. For example, the time release studies advocated by the World Customs Organisation (WCO) measure the time to release the import cargo from regulatory purview (WCO 2018). However, being at the macro level such measures do not point to the root causes of delays in the release of cargo from customs purview. Comparatively, the current study prompts the collection of granular temporal data at process and activity levels. When temporal data is depicted alongside the process map as in the RAD-based mapping formalism, segments consuming comparatively longer times are easily identified, pointing to potential process issues. Furthermore, the reference model is able to identify segments that lack detailed temporal data prompting deployment of technologies to collect them. For example, instead of looking at truck-turn-around data at a container terminal which is based on entry and exit times for a truck, granular data that reveals problems along the journey can be gathered through technologies such as Radio Frequency Identification (RFID), Global Positioning Systems (GPS) or Wireless Sensor Networks (WSN) (Thennakoon et al. 2024). The reference model also points to possibilities of gathering granular data for informational processes through technologies such as process mining. These technologies are included in the reference model for each dimension, which provides further insights into processes paving the way for continual process improvements.

Considering the multi-stakeholder nature of CBL operations, interfaces among stakeholders play a critical role in CBL performance, a concept reinforced in the supply chain integration literature (Williams et al. 2013). Grainger and Morini (2019), for example, have classified the interfaces between regulators and users in CBL into three different categories and made recommendations to improve each. The reference model, which treats interactions as a separate process dimension, draws attention to interfaces among all stakeholders within a CBL system. This analysis can be further extended to classify interfaces based on their duration and the amount of information exchanged to further explicate the type of interface management required (Bandaranayake, Kiridena, Kulatunga, & Dam, 2022c). The prominence provided to the interactional dimension in the reference model eliminates the need for a separate formalism such as service blueprinting (Wilson et al. 2016) to study customer perceptions. Furthermore, the analysis through the interactional dimension provides guidance on operationalising supply chain integration by identifying technologies for different types of interfaces encountered in CBL. For example, daily interactions between a haulage company and

a container terminal can be reduced by the one-time update of fleet details with the terminal and re-using such data at each entry and exit.

Treatment of spatial dimension as a distinct dimension aid in identifying opportunities to improve processes through the deployment of technologies such as IoT sensors, RFID and cameras. As demonstrated through the instance of reading container numbers in loading containers to trains at the intermodal terminal, the reference model provides the guidance to embed these technologies seamlessly into processes. The reference model also emphasises the connectivity of the spatial dimension to the rest of the system. When such linkages between the physical system and the digital systems are well-implemented, it paves the way to the creation of digital twins (Klar et al. 2023) enabling real-time visualisation of the physical system and the control of the physical system through digital interfaces. As such, the reference model is a useful tool in moving towards the implementation of digital twins which has the potential to improve CBL systems substantially.

Applicable technologies have been grouped logically in the process dimensions. For example, the Internet of Things (IoT) devices, sensors, and RFID all fit into the spatial dimension and the technologies that mimic cognitive tasks such as AI are grouped in the control dimension. Such classification provides clarity in assessing the technologies to adopt in different operations and processes and goes beyond the mere identification of applicable technologies (Heilig and Voß 2017). How these different technologies could be combined to achieve synergistic improvements can also be studied, aided by the model, leading to the development of a technology roadmap to facilitate investment decisions (Bandaranayake, Kiridena, & Kulatunga, 2022a). Furthermore, the reference model provides a means for deconstructing jobs into constituent activities to identify automation opportunities as suggested by Jesuthasan and Boudreau (2018).

From a technology management perspective, monolithic solutions such as National Single Windows (NSWs) owned or managed by a single stakeholder (usually a government agency) are prone to obsolescence and difficulties of maintenance. There are parallels to the adoption of Enterprise Resource Planning (ERP) systems, which promised to be a unified solution for a multitude of organisational applications but resulted in many claims of unrealized return on investment (Sammon and Adam 2007). The reference model takes a more nuanced view of technological solutions segmenting them into various process dimensions and proposing a modular approach. This enables the selection of technology components suited for each dimension of each stakeholder's operations. For example, appropriately designed interfaces based on service-oriented architectures (SOA) can serve to reduce the burden of multiple bilateral

interfaces, which is easier to manage compared to the implementation of a single system. The Australian system offers an example of the implementation of such a system through a message broker. This is an example of the capabilities of the reference model to identify best practices.

Performance benchmarking as offered by the Logistics Performance Index (LPI) of the World Bank offers a means of comparing logistics performance among different countries. Being the outcome of a perception-based survey, LPI carries some subjectivity. Furthermore, it does not provide clarity on what to improve in a CBL system. Comparatively, the process benchmarking approach demonstrated through the study offers an objective approach to benchmarking, which also points to the processes to improve and technologies to deploy.

The analysis through the framework developed in the study sought to provide a uniform coverage of the CBL chain affording attention to both logistical and regulatory aspects. This is a departure from the focus on selected segments, such as the concentration of studies on terminal operations, as highlighted by Fransoo and Lee (2013). The study also heeds the call for a supply chain orientation in the CBL domain as opposed to the asset utilisation orientation of shipping lines and terminal operators which, leads to sub-optimisations.

The applicability of the reference model to widely varying systems has been demonstrated through the analysis of contrasting CBL systems. The scalability of the reference model means that it can be applied both at the macro level and at the micro level. For example, bodies such as national trade facilitation committees can make use of the tool to evaluate the feasibility and architecture of systems such as inter-organisational information systems or intermodal logistics hubs to suit the prevailing systems. Similarly, individual stakeholders can use the tools to conceptualize their improvement efforts within the broader context of the CBL systems and select technologies that would be compatible with the system.

The reference model requires the processes to be captured at sufficient granularity to be analysed, which requires significant effort. While this may appear as a limitation, considering the potential gains from improvements, resulting in cost and time savings, the investment in process discovery is justified. Once captured, the process maps can be updated with relatively less effort as processes evolve. The reference model may appear limited in its ability to evaluate the practicality of the identified improvements. However, as identified through the benchmarking approach, some of the identified solutions have already been deployed in other systems providing evidence of the validity of the approach. The development of a simulation approach based on process dimensions can be undertaken as part of future research

to further enhance the applicability of the framework. The application of the framework to related systems such as air-cargo systems and land border crossings are further potential research directions.

7 Conclusion

CBL operations exhibit characteristics from both product and service delivery systems ranging from highly repeatable to case-specific activities. Further confounded by the involvement of a multitude of stakeholders from a range of industries and government agencies, CBL requires an improvement approach catered to its inherent properties for analysis and improvement, a gap fulfilled by this study. The reference model proposed in this paper aims to address the challenges in CBL by positioning it at the meso-level; i.e. between macro-level solutions proposed at the broader system level and micro-level optimizations applied to specific segments of operations.

This paper builds upon the fundamental notion of process dimensions by proposing a formalism to capture CBL operations integrating seven process dimensions and leveraging them to analyse and improve those processes. The recognition of the prominence of spatial, temporal and interactional dimensions is a novel contribution to process analysis traditionally based on functional, organizational, control and informational dimensions.

The study reported in this paper explored two distinct CBL systems with contrasting characteristics in terms of geography, type of operations and the application of technology. The detailed process maps developed in the study are a key contribution as they not only provide holistic and unified insights into each CBL system but also guide similar studies in CBL systems elsewhere. The study offers a reference model that can be readily applied in practice. The elegance of the reference model lies in its simplicity of application. However, it possesses the requisite depth to compel practitioners to think differently compared to when using existing methodologies for process improvement. It also provides insights into the additional process dimensions that lead to improved operations. The proposed reference model was applied to selected segments of operations in each case, demonstrating its capabilities to not only analyse processes but also benchmark different systems. The reference model offers a smooth progression from process modelling to analysis and improvement in place of the disparate suite of tools and methodologies currently used throughout the process management lifecycle. Furthermore, the reference model facilitates the search for appropriate technological solutions in each segment of CBL operations. The RAD-based modelling formalism and the reference

model constitute a valuable contribution to practice in process analysis and improvement.

From a theoretical perspective, the process dimension-based analysis is a useful contribution to process management literature. It not only offers a novel line of inquiry into process analysis through the lens of process dimensions but also for classifying technology into those dimensions. Each process dimension opens up a distinct line of inquiry that offers depth to the analysis. The approach provides a means of integrating technology into processes, by way of tightly coupling processes and technologies leading to improved process performance.

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Data availability The data that support the findings of this study are available on request from the corresponding author. The data are not made publicly available due to containing information that could compromise the privacy of research participants.

Declarations

Ethics approval This research has been approved by the University of Wollongong and Illawarra and Shoalhaven Local Health District Social Sciences Human Research Ethics Committee (Approval Number: 2021/455).

Conflict of interest The authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest, or non-financial interest in the subject matter or materials discussed in this manuscript.

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